

PRACTICAL – 10

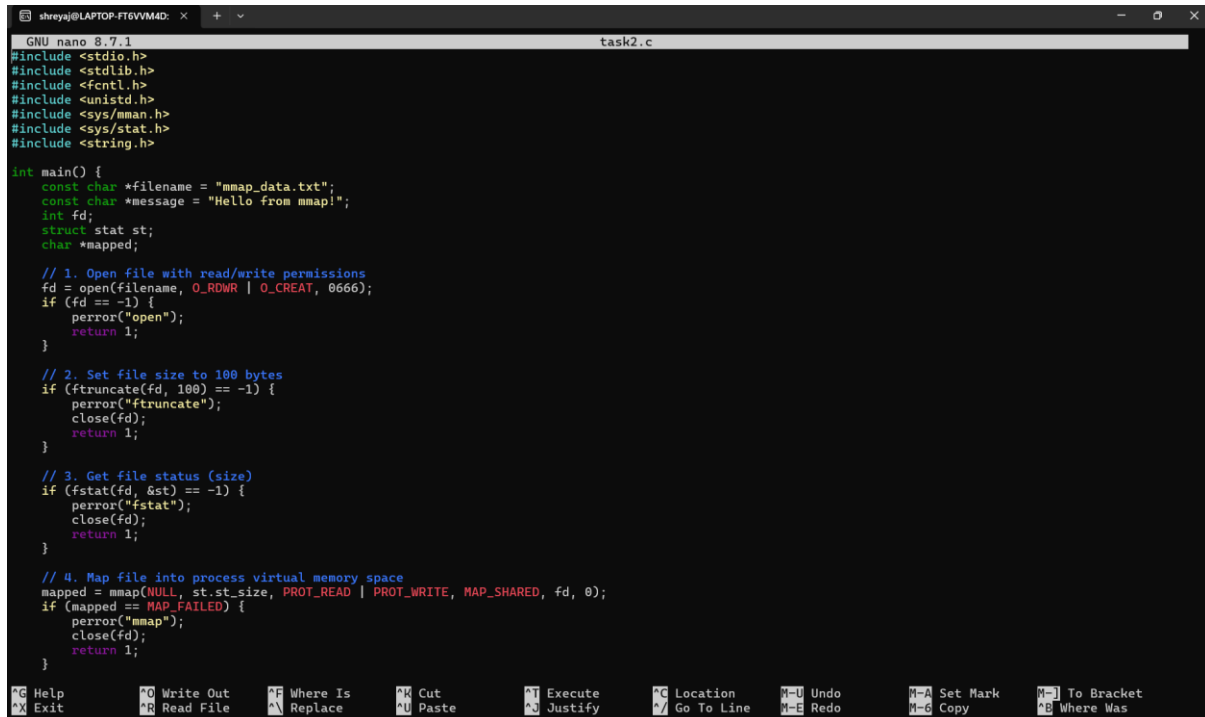
TASK1: Investigate inode structures using commands such as `ls -li`, `stat`, and `find`. Create hard links and symbolic links and explain their impact on inode allocation.

```
shreyaj@LAPTOP-FT6VVM4D: ~$ mkdir -p ~/Practical10
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cd ~/Practical10
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ echo "Hello Ubuntu" > original.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ stat original.txt
  File: original.txt
  Size: 13          Blocks: 8          IO Block: 4096   regular file
Device: 8:48      Inode: 56601         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyaj)   Gid: ( 1000/ shreyaj)
Access: 2026-09-28 03:38:41.238977427 +0000
Modify: 2026-09-28 03:38:41.238977427 +0000
Change: 2026-09-28 03:38:41.238977427 +0000
Birth: 2026-09-28 03:38:41.238977427 +0000
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ ln original.txt hardlink.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ ls -li original.txt hardlink.txt
56601 hardlink.txt 56601 original.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ echo "New line" >> hardlink.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cat original.txt
Hello Ubuntu
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cat hardlink.txt
Hello Ubuntu
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ ln -s original.txt symlink.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ ls -li
total 4
-rw-r--r-- 1 shreyaj shreyaj 13 2026-09-28 03:38 original.txt
-rw-r--r-- 1 shreyaj shreyaj 13 2026-09-28 03:38 hardlink.txt
lrwxrwxrwx 1 shreyaj shreyaj 12 2026-09-28 03:38 symlink.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ stat -L symlink.txt
  File: symlink.txt -> 'original.txt'
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8:48      Inode: 56611         Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/ shreyaj)   Gid: ( 1000/ shreyaj)
Access: 2026-09-28 03:39:07.698976342 +0000
Modify: 2026-09-28 03:39:07.698976341 +0000
Change: 2026-09-28 03:39:07.698976341 +0000
Birth: 2026-09-28 03:39:07.698976341 +0000
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ stat -L symlink.txt
  File: symlink.txt
  Size: 22          Blocks: 8          IO Block: 4096   regular file
Device: 8:48      Inode: 56601         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyaj)   Gid: ( 1000/ shreyaj)
Access: 2026-09-28 03:39:00.442977380 +0000
Modify: 2026-09-28 03:39:00.430977383 +0000
Change: 2026-09-28 03:39:00.430977383 +0000
Birth: 2026-09-28 03:38:41.238977427 +0000
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ find . -inum 50825
.
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ rm original.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ ls -li
total 4
-rw-r--r-- 1 shreyaj shreyaj 13 2026-09-28 03:38 hardlink.txt
lrwxrwxrwx 1 shreyaj shreyaj 12 2026-09-28 03:38 symlink.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cat hardlink.txt
Hello Ubuntu
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cat symlink.txt
Hello Ubuntu
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$
```

```
shreyaj@LAPTOP-FT6VVM4D: ~$ stat original.txt
  File: original.txt
  Size: 13          Blocks: 8          IO Block: 4096   regular file
Device: 8:48      Inode: 56601         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyaj)   Gid: ( 1000/ shreyaj)
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56601 hardlink.txt 56601 original.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ echo "New line" >> hardlink.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cat original.txt
Hello Ubuntu
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cat hardlink.txt
Hello Ubuntu
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ ln -s original.txt symlink.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ ls -li
total 4
-rw-r--r-- 1 shreyaj shreyaj 13 2026-09-28 03:38 original.txt
-rw-r--r-- 1 shreyaj shreyaj 13 2026-09-28 03:38 hardlink.txt
lrwxrwxrwx 1 shreyaj shreyaj 12 2026-09-28 03:38 symlink.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ stat -L symlink.txt
  File: symlink.txt -> 'original.txt'
  Size: 12          Blocks: 0          IO Block: 4096   symbolic link
Device: 8:48      Inode: 56611         Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/ shreyaj)   Gid: ( 1000/ shreyaj)
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Access: (0644/-rw-r--r--)  Uid: ( 1000/ shreyaj)   Gid: ( 1000/ shreyaj)
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shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ find . -inum 50825
.
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ rm original.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ ls -li
total 4
-rw-r--r-- 1 shreyaj shreyaj 13 2026-09-28 03:38 hardlink.txt
lrwxrwxrwx 1 shreyaj shreyaj 12 2026-09-28 03:38 symlink.txt
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cat hardlink.txt
Hello Ubuntu
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cat symlink.txt
Hello Ubuntu
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$ cat: symlink.txt: No such file or directory
shreyaj@LAPTOP-FT6VVM4D: ~/Practical10$
```

TASK2: Design a program using `mmap()` to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity.

```
shreyaj@LAPTOP-FT6VVM4D:~/Practical10$ nano task2.c
shreyaj@LAPTOP-FT6VVM4D:~/Practical10$ gcc task2.c -o task2
./task2
Data written using mmap(): Hello from mmap!
Data read using mmap(): Hello from mmap!
shreyaj@LAPTOP-FT6VVM4D:~/Practical10$ cat mmap_data.txt
shreyaj@LAPTOP-FT6VVM4D:~/Practical10$ |
```



```
GNU nano 2.7.1 task2.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main() {
    const char *filename = "mmap_data.txt";
    const char *message = "Hello from mmap!";
    int fd;
    struct stat st;
    char *mapped;

    // 1. Open file with read/write permissions
    fd = open(filename, O_RDWR | O_CREAT, 0666);
    if (fd == -1) {
        perror("open");
        return 1;
    }

    // 2. Set file size to 100 bytes
    if (ftruncate(fd, 100) == -1) {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    // 3. Get file status (size)
    if (fstat(fd, &st) == -1) {
        perror("fstat");
        close(fd);
        return 1;
    }

    // 4. Map file into process virtual memory space
    mapped = mmap(NULL, st.st_size, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);
    if (mapped == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    // 5. Write data into memory buffer directly
    strcpy(mapped, message);

    // 6. Flush changes from virtual memory to physical disk
    if (msync(mapped, st.st_size, MS_SYNC) == -1) {
        perror("msync");
    }

    // 7. Read and verify written data
    printf("Data written using mmap(): %s\n", mapped);
    printf("Data read using mmap(): %s\n", mapped);

    // 8. Unmap memory and close file descriptor
    if (munmap(mapped, st.st_size) == -1) {
        perror("munmap");
    }

    close(fd);
    return 0;
}
```

```
// 5. Write data into memory buffer directly
strcpy(mapped, message);

// 6. Flush changes from virtual memory to physical disk
if (msync(mapped, st.st_size, MS_SYNC) == -1) {
    perror("msync");
}

// 7. Read and verify written data
printf("Data written using mmap(): %s\n", mapped);
printf("Data read using mmap(): %s\n", mapped);

// 8. Unmap memory and close file descriptor
if (munmap(mapped, st.st_size) == -1) {
    perror("munmap");
}

close(fd);
return 0;
}
```